

Conservative oscillations and perturbation

Teaching sequence

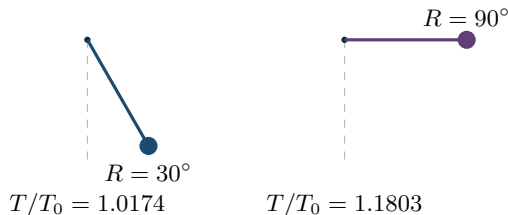
- 2.1 Energy and amplitude-dependent period
- 2.2 Backbone curves
- 2.3 Regular perturbation and secular terms
- 2.4 Poincaré–Lindstedt: cubic stiffness
- 2.5 Quadratic stiffness and mean shift
- 2.6 Competing quadratic and cubic effects
- 2.7 Validation, limitations and solved practice

A periodic motion can remain bounded while its frequency and waveform change with amplitude. This week: predict both and check the approximation.

2.1 From energy to amplitude-dependent period

Does a larger pendulum swing take longer?

Two releases from rest have the same L and g , but different initial angles. The ratios below come from the exact ideal-pendulum period integral derived next.



The larger release takes longer. The linear period $T_0 = 2\pi\sqrt{L/g}$ has no amplitude in it, so it cannot explain this difference.

Conserved energy relates the release angle R to the time taken for one swing.

2.1 From energy to amplitude-dependent period

Energy gives the period without solving the motion

$$m\ddot{x} + \frac{dU}{dx} = 0, \quad E = \frac{1}{2}m\dot{x}^2 + U(x).$$

Multiply the equation by $\dot{x}$: $m\ddot{x}\dot{x} + (dU/dx)\dot{x} = dE/dt = 0$.

The turning points x_- and x_+ satisfy $U(x_-) = U(x_+) = E$. Between them $E - U(x) > 0$, so the velocity is real.

On the rightward branch

$$\dot{x} = \sqrt{\frac{2}{m}[E - U(x)]}, \quad dt = \sqrt{\frac{m}{2}} \frac{dx}{\sqrt{E - U(x)}}.$$

Travel from x_- to x_+ takes half a cycle. Double its time integral.

$$T(E) = \sqrt{2m} \int_{x_-}^{x_+} \frac{dx}{\sqrt{E - U(x)}}, \quad \omega(E) = \frac{2\pi}{T(E)}.$$

2.1 From energy to amplitude-dependent period

Worked example: recover the linear period

$$U(x) = \frac{1}{2}kx^2, \quad E = \frac{1}{2}kR^2, \quad x_{\pm} = \pm R.$$

$$T = 2\sqrt{m/k} \int_{-R}^R \frac{dx}{\sqrt{R^2 - x^2}}.$$

With $x = R \sin \psi$, the limits become $-\pi/2, \pi/2$ and $dx/\sqrt{R^2 - x^2} = R \cos \psi \, d\psi / (R \cos \psi) = d\psi$.

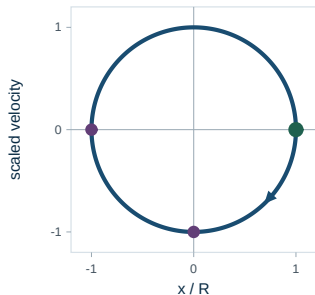
$$T = 2\sqrt{m/k} \int_{-\pi/2}^{\pi/2} d\psi = \boxed{2\pi\sqrt{m/k}}.$$

The amplitude cancels. Thus $\omega_0 = \sqrt{k/m}$ is independent of R for this linear model.

2.1 From energy to amplitude-dependent period

A closed phase orbit takes one period

For the linear spring, $x/R = \cos(\omega_0 t)$ and $v/(\omega_0 R) = -\sin(\omega_0 t)$, where $v = \dot{x}$. This recalls the Week 1 phase portrait and adds travel time.



Release: $x = R, v = 0$

First quarter: $x = 0$

Half-cycle: $x = -R$

Return to R : one period T

Time runs clockwise.

Starting at $(x, v) = (R, 0)$, the orbit reaches $x = 0$ after $T/4$ and $x = -R$ after $T/2$. One closed lap takes T , the time measured by the energy integral.

2.1 From energy to amplitude-dependent period

Pendulum: release energy gives the angular speed

$$\theta(0) = R, \quad \dot{\theta}(0) = 0, \quad 0 < R < \pi, \quad \omega_0 = \sqrt{g/L}.$$

The model is $\ddot{\theta} + \omega_0^2 \sin \theta = 0$. Multiplication by $\dot{\theta}$ gives a total time derivative:

$$\frac{d}{dt} \left[\frac{1}{2} \dot{\theta}^2 + \omega_0^2 (1 - \cos \theta) \right] = 0.$$

At release $\dot{\theta} = 0$, so the conserved quantity equals $\omega_0^2(1 - \cos R)$:

$$\frac{1}{2} \dot{\theta}^2 + \omega_0^2 (1 - \cos \theta) = \omega_0^2 (1 - \cos R).$$

$$\dot{\theta}^2 = 2\omega_0^2 (\cos \theta - \cos R).$$

On the first quarter-cycle θ decreases from R to 0. Use the negative square root for $\dot{\theta}$ when converting $dt = d\theta/\dot{\theta}$.

2.1 From energy to amplitude-dependent period

Pendulum: substitute into the quarter-period integral

During the first quarter-cycle θ falls from R to 0, so $\dot{\theta} = -\omega_0 \sqrt{2(\cos \theta - \cos R)}$. With $dt = d\theta/\dot{\theta}$, reverse the limits to make the time positive:

$$\begin{aligned}\frac{T}{4} &= \int_R^0 \frac{d\theta}{-\omega_0 \sqrt{2(\cos \theta - \cos R)}} \\ \frac{T}{4} &= \frac{1}{\omega_0} \int_0^R \frac{d\theta}{\sqrt{2(\cos \theta - \cos R)}}.\end{aligned}$$

This is the time for one quarter of the orbit. Symmetry gives four equal quarters.

2.1 From energy to amplitude-dependent period

Pendulum: transform the period integral

Start from $T = 4\omega_0^{-1} \int_0^R [2(\cos \theta - \cos R)]^{-1/2} d\theta$.

Set $\sin(\theta/2) = \kappa \sin \psi$, where $\kappa = \sin(R/2)$. The new limits are 0 and $\pi/2$.

$$2(\cos \theta - \cos R) = 4\kappa^2 \cos^2 \psi, \quad d\theta = \frac{2\kappa \cos \psi}{\sqrt{1 - \kappa^2 \sin^2 \psi}} d\psi.$$

Cancel $2\kappa \cos \psi$ between numerator and denominator. Four equal quarter-cycles give T .

$$T = \frac{4}{\omega_0} K(\kappa), \quad K(\kappa) = \int_0^{\pi/2} \frac{d\psi}{\sqrt{1 - \kappa^2 \sin^2 \psi}}.$$

κ is the elliptic modulus; some software instead accepts the parameter κ^2 . The integral is exact for the ideal pendulum.

2.1 From energy to amplitude-dependent period

Pendulum: expand the period at small amplitude

For small release angle R , the modulus $\kappa = \sin(R/2)$ is small. Expand the integrand of $K(\kappa)$ by setting $z = \kappa^2 \sin^2 \psi$ in the binomial series:

$$(1 - z)^{-1/2} = 1 + \frac{1}{2}z + \frac{3}{8}z^2 + \cdots, \quad K = \frac{\pi}{2} \left(1 + \frac{\kappa^2}{4} + \frac{9\kappa^4}{64} + \cdots \right).$$

Integrate powers term by term: $\int_0^{\pi/2} \sin^2 \psi \, d\psi = \pi/4$ and $\int_0^{\pi/2} \sin^4 \psi \, d\psi = 3\pi/16$.

Use $\kappa^2 = R^2/4 - R^4/48 + O(R^6)$ and $\kappa^4 = R^4/16 + O(R^6)$; collect equal powers of R .

$$\boxed{\frac{T}{T_0} = 1 + \frac{R^2}{16} + \frac{11R^4}{3072} + O(R^6)}, \quad T_0 = \frac{2\pi}{\omega_0}.$$
$$\frac{\omega}{\omega_0} = 1 - \frac{R^2}{16} + \frac{R^4}{3072} + O(R^6).$$

As R increases within the range where this small-angle series is accurate, the period increases and the frequency decreases. The exact integral shows the same trend at larger angles.

2.1 From energy to amplitude-dependent period

Pendulum: calculate and compare

For $L = 1$ m and $g = 9.81$ m/s², $T_0 = 2.006067$ s.

R	Exact T/T_0	Exact T (s)	$1 + R^2/16$
30°	1.017409	2.040990	1.017135
60°	1.073182	2.152875	1.068539
90°	1.180341	2.367842	1.154213

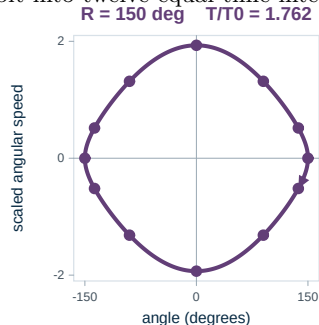
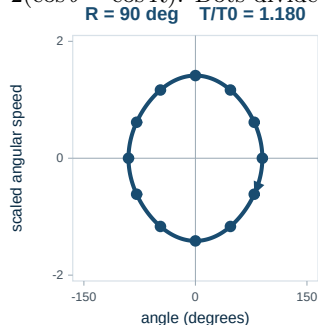
At 60°, convert to $R = \pi/3$ rad before using $1 + R^2/16 = 1.068539$.

$T \approx 2.006067(1.068539) = 2.14356$ s, about 0.433% below the exact ideal-pendulum period 2.152875 s. The leading correction is useful here, but its error grows with R .

2.1 From energy to amplitude-dependent period

A larger pendulum orbit takes a longer lap

For each release, $(\dot{\theta}/\omega_0)^2 = 2(\cos \theta - \cos R)$. Dots divide each orbit into twelve equal time intervals.



The 150° swing moves faster at $\theta = 0$ yet takes $1.762T_0$ per lap, versus $1.180T_0$ at 90°. Near $R = \pi$, slowing by the turning angle makes the period grow sharply.

2.2 Backbone curves

One free orbit gives one backbone point

Release from rest at $x(0) = R$: follow that orbit to its frequency

$$R = x_+ \longrightarrow E = U(R) \longrightarrow T(E) \longrightarrow (R, \omega), \quad \omega = \frac{2\pi}{T(E)}.$$

Repeat for several releases R and plot the resulting points (R, ω) . The curve through them is the free-oscillation backbone. Here its amplitude axis means release displacement R .

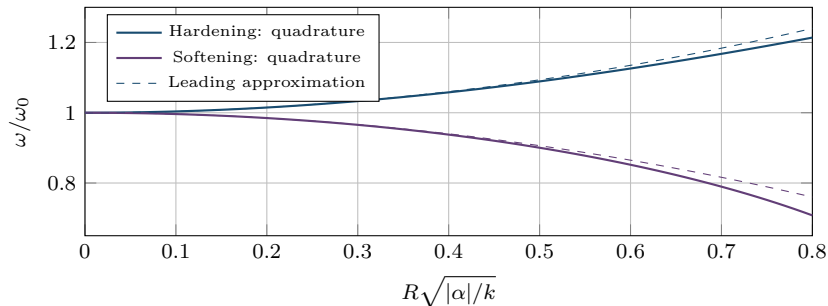
$$\frac{d\omega}{dR} > 0 : \text{hardening}, \quad \frac{d\omega}{dR} < 0 : \text{softening}.$$

Here $R = x(0) > 0$ is the release displacement and $\dot{x}(0) = 0$. The backbone plots frequency against this R .

2.2 Backbone curves

Hardening and softening backbones

For $m\ddot{x} + kx + \alpha x^3 = 0$, the horizontal axis is the release R scaled by $\sqrt{|\alpha|/k}$. Positive α hardens; negative α softens.



Solid: period from the energy integral. Dashed: the small-amplitude approximation derived in 2.4. At the same release R , positive α raises the frequency and negative α lowers it.

2.2 Backbone curves

Cubic spring: period from conserved energy

Conserved energy gives a period for the cubic-spring model at any bounded release.

$$m\ddot{x} + kx + \alpha x^3 = 0, \quad x = Ru, \quad \tau = \sqrt{k/m} t, \quad \varepsilon = \alpha R^2/k.$$

$$u'' + u + \varepsilon u^3 = 0, \quad u(0) = 1, \quad u'(0) = 0.$$

Multiply the equation by u' and integrate. At release $u = 1$, $u' = 0$, so

$$\frac{1}{2}u'^2 + \frac{1}{2}u^2 + \frac{\varepsilon}{4}u^4 = \frac{1}{2} + \frac{\varepsilon}{4}.$$

Rearrange and factor the velocity squared:

$$u'^2 = (1 - u^2) \left[1 + \frac{\varepsilon}{2}(1 + u^2) \right].$$

On the first quarter-cycle u decreases from 1 to 0. The energy equation gives its speed without solving for $u(\tau)$.

2.2 Backbone curves

Turn the cubic-spring speed into a period

On the first quarter-cycle take the negative root of u'^2 and use $d\tau = du/u'$. Reversing the limits gives

$$\frac{T_\tau}{4} = \int_0^1 \frac{du}{\sqrt{(1-u^2)[1+\frac{\varepsilon}{2}(1+u^2)]}}.$$

Set $u = \sin \psi$, so $du = \cos \psi d\psi$ and $\sqrt{1-u^2} = \cos \psi$ for $0 \leq \psi \leq \pi/2$. They cancel.

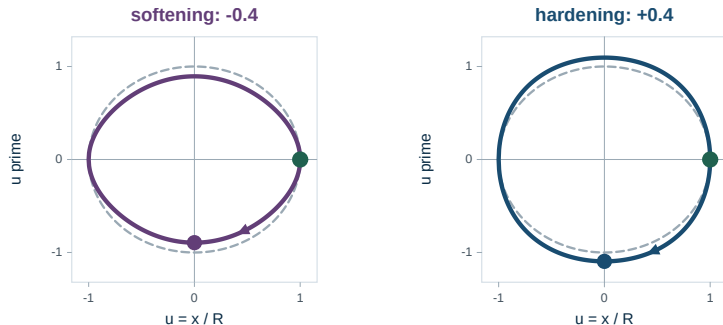
$$\boxed{T_\tau = 4 \int_0^{\pi/2} \frac{d\psi}{\sqrt{1+\frac{\varepsilon}{2}(1+\sin^2 \psi)}}}, \quad \Omega = \frac{\omega}{\omega_0} = \frac{2\pi}{T_\tau}.$$

This integral is exact for the stated cubic-spring model and supplies an independent check of the perturbation approximation.

2.2 Backbone curves

At the same release, cubic stiffness changes the orbit

For $u'' + u + \varepsilon u^3 = 0$, compare $\varepsilon = -0.4$ and $+0.4$ at the same release $u = 1$ from rest. The dashed loop is the linear case.

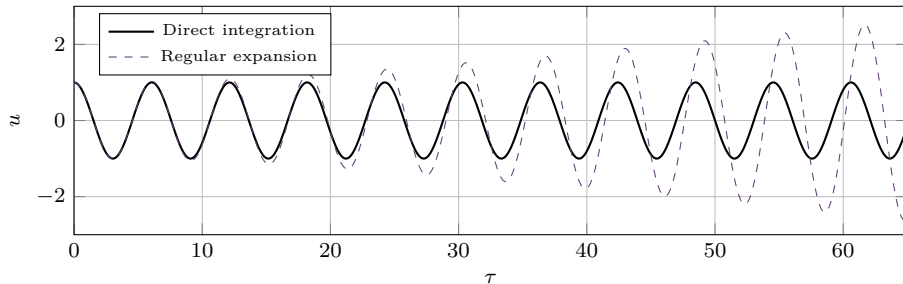


Energy gives $|u'| = \sqrt{1 + \varepsilon/2}$ at $u = 0$: 0.894 for $\varepsilon = -0.4$ and 1.095 for $+0.4$. The full-orbit integral gives $T/T_0 = 1.200$ and 0.878, respectively.

2.3 Regular perturbation and secular terms

Can a bounded oscillator have a growing approximation?

$$u'' + u + \varepsilon u^3 = 0, \quad \varepsilon = 0.10, \quad u(0) = 1, \quad u'(0) = 0.$$



$\varepsilon = 0.10$ and release $(u, u') = (1, 0)$. The first-order approximation drifts because it keeps the linear frequency; the coming method corrects that frequency.

2.3 Regular perturbation and secular terms

Expand the displacement at fixed linear time

$$u'' + u + \varepsilon u^3 = 0, \quad u(0) = 1, \quad u'(0) = 0, \quad |\varepsilon| \ll 1.$$

$$u = u_1 + \varepsilon u_2 + O(\varepsilon^2).$$

Subscripts number successive terms. Since $u = x/R$ already has unit release, its leading term is u_1 . Substitute $\varepsilon u^3 = \varepsilon u_1^3 + O(\varepsilon^2)$ and match powers of ε :

$$O(1): \quad u_1'' + u_1 = 0, \quad u_1(0) = 1, \quad u_1'(0) = 0,$$

$$O(\varepsilon): \quad u_2'' + u_2 = -u_1^3, \quad u_2(0) = u_2'(0) = 0.$$

$u_1 = \cos \tau$. Primes mean $d/d\tau$, where $\tau = \omega_0 t$ and $\omega_0 = \sqrt{k/m}$.

2.3 Regular perturbation and secular terms

Solve the equation for the next term

$$u_2'' + u_2 = -\cos^3 \tau = -\frac{3}{4} \cos \tau - \frac{1}{4} \cos 3\tau.$$

The right-hand side has two cosines. To solve for each part, direct differentiation gives

$$(\cos 3\tau)'' + \cos 3\tau = -8 \cos 3\tau, \quad (\tau \sin \tau)'' + \tau \sin \tau = 2 \cos \tau.$$

Therefore a particular solution is $-3\tau \sin \tau/8 + \cos 3\tau/32$. Add $C \cos \tau + S \sin \tau$. The conditions $u_2(0) = u_2'(0) = 0$ give $C = -1/32$ and $S = 0$.

$$u_2 = -\frac{3}{8} \tau \sin \tau + \frac{\cos 3\tau - \cos \tau}{32}.$$

The resonant term $-3\tau \sin \tau/8$ is secular. Its contribution becomes order one when $|\varepsilon|\tau = O(1)$. This is approximation failure, not physical instability.

2.3 Regular perturbation and secular terms

A small frequency shift creates the secular term

The term $-3\varepsilon\tau \sin \tau/8$ in εu_2 comes from expanding a cosine with a slightly changed frequency:

$$\cos \left[\left(1 + \frac{3\varepsilon}{8} \right) \tau \right] = \cos \tau \cos \frac{3\varepsilon\tau}{8} - \sin \tau \sin \frac{3\varepsilon\tau}{8}.$$

When $|\varepsilon\tau| \ll 1$, use $\cos(3\varepsilon\tau/8) \simeq 1$ and $\sin(3\varepsilon\tau/8) \simeq 3\varepsilon\tau/8$. This reproduces $\cos \tau - 3\varepsilon\tau \sin \tau/8$.

For $\varepsilon = 0.10$, ten linear periods mean $\tau = 20\pi$. The first-order phase change is

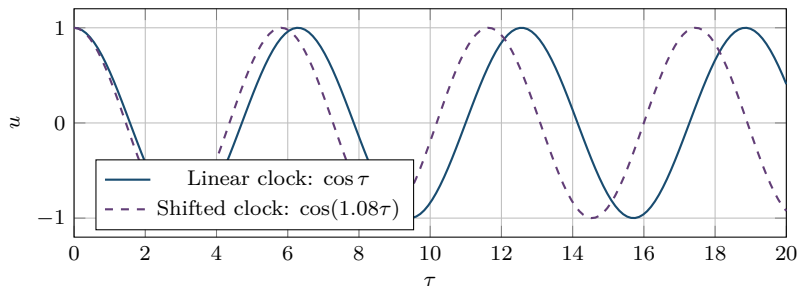
$$\tau = 20\pi, \quad \Delta\phi \approx \frac{3\varepsilon}{8}\tau = \boxed{2.35619 \text{ rad} = 135^\circ}.$$

This large $\Delta\phi$ is beyond the small-phase expansion. It shows why the truncated regular series loses accuracy after many cycles; the motion itself remains bounded.

2.4 Poincaré–Lindstedt method for cubic stiffness

The oscillator needs a corrected clock

The motion stays bounded, but a small frequency error grows into a visible phase error. The 8% shift below illustrates the idea; it is not a fitted model result.



After time τ , the phase difference is 0.08τ . Poincaré–Lindstedt expands the frequency along with the waveform, so the approximation uses the corrected clock.

2.4 Poincaré–Lindstedt method for cubic stiffness

Use a phase that follows the unknown frequency

In $u'' + u + \varepsilon u^3 = 0$, primes mean derivatives in $\tau = \omega_0 t$. Describe the motion using the new phase $\phi = \Omega\tau$, where $\Omega = \omega/\omega_0$ is unknown. Expand it as

$$\Omega = 1 + \varepsilon\Omega_1 + O(\varepsilon^2).$$

The chain rule gives $d\phi/d\tau = \Omega$. Since Ω is constant for a given motion,

$$\frac{du}{d\tau} = \Omega u_\phi, \quad \frac{d^2u}{d\tau^2} = \Omega^2 u_{\phi\phi}.$$

$$\boxed{\Omega^2 u_{\phi\phi} + u + \varepsilon u^3 = 0}, \quad \Omega^2 = 1 + 2\varepsilon\Omega_1 + O(\varepsilon^2).$$

One cycle is 2π in ϕ , so its dimensionless period is $T_\tau = 2\pi/\Omega$. The unknown Ω_1 can remove the secular term.

2.4 Poincaré–Lindstedt method for cubic stiffness

Collect the first-order terms

Insert $u = u_1 + \varepsilon u_2 + \cdots$ and $\Omega^2 = 1 + 2\varepsilon\Omega_1 + \cdots$ into $\Omega^2 u_{\phi\phi} + u + \varepsilon u^3 = 0$.

$$O(1): \quad u_{1,\phi\phi} + u_1 = 0, \quad u_1(0) = 1, \quad u_{1,\phi}(0) = 0 \quad \implies \quad u_1 = \cos \phi.$$

At $O(\varepsilon)$, the acceleration contributes $u_{2,\phi\phi} + 2\Omega_1 u_{1,\phi\phi}$. The spring terms contribute $u_2 + u_1^3$. Thus

$$u_{2,\phi\phi} + u_2 = -u_1^3 - 2\Omega_1 u_{1,\phi\phi}.$$

Use $u_{1,\phi\phi} = -\cos \phi$ and $\cos^3 \phi = (3 \cos \phi + \cos 3\phi)/4$:

$$u_{2,\phi\phi} + u_2 = \left(2\Omega_1 - \frac{3}{4}\right) \cos \phi - \frac{1}{4} \cos 3\phi.$$

2.4 Poincaré–Lindstedt method for cubic stiffness

Remove the resonant term and satisfy release data

A $\cos \phi$ term on the right would produce $\phi \sin \phi$. Set its coefficient to zero: $2\Omega_1 - 3/4 = 0$. Then

$$\boxed{\Omega_1 = \frac{3}{8}}, \quad u_{2,\phi\phi} + u_2 = -\frac{1}{4} \cos 3\phi.$$

The $\cos 3\phi$ part gives $\cos 3\phi/32$. Add a free $\cos \phi$ term so $u_2(0) = 0$; the zero initial velocity removes the sine term.

$$\boxed{u_2 = \frac{\cos 3\phi - \cos \phi}{32}.}$$
$$u \approx \cos \phi + \frac{\varepsilon}{32}(\cos 3\phi - \cos \phi), \quad \phi = \left(1 + \frac{3\varepsilon}{8}\right) \tau.$$

This approximation stays periodic. The correction is zero at $\phi = 0$, so it preserves release from $u(0) = 1$.

2.4 Poincaré–Lindstedt method for cubic stiffness

Worked example: a hardening spring

$$m = 1 \text{ kg}, \quad k = 100 \text{ N/m}, \quad \alpha = 1000 \text{ N/m}^3, \quad R = 0.10 \text{ m}.$$

First find the linear frequency and the dimensionless nonlinear strength at the stated release:

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{100}{1}} = 10 \text{ rad/s}, \quad \varepsilon = \frac{\alpha R^2}{k} = \frac{1000(0.10)^2}{100} = 0.10.$$

Then apply $\Omega = 1 + 3\varepsilon/8$ and convert back to physical time:

$$\Omega_{\text{PL}} = 1 + \frac{3(0.10)}{8} = 1.0375, \quad \omega_{\text{PL}} = \omega_0 \Omega_{\text{PL}} = 10(1.0375) = 10.375 \text{ rad/s}.$$

$$T_{\text{PL}} = \frac{2\pi}{\omega_{\text{PL}}} = \frac{2\pi}{10.375} = 0.605608 \text{ s}.$$

Exact-model quadrature: $\omega = 10.36716907 \text{ rad/s}$ and $T = 0.606065674 \text{ s}$. The first-order frequency error is 0.075536%.

2.4 Poincaré–Lindstedt method for cubic stiffness

Hardening example: check the harmonic amplitudes

From the preceding result, $\phi = \omega_{\text{PL}}t = 10.375t$ and $\varepsilon/32 = 0.10/32 = 0.003125$.

$$\begin{aligned}x(t) &\approx 0.10 [\cos(10.375t) + 0.003125\{\cos(31.125t) - \cos(10.375t)\}] \text{ m}, \\ &= 0.0996875 \cos(10.375t) + 0.0003125 \cos(31.125t) \text{ m}.\end{aligned}$$

The fundamental component is slightly smaller than the release displacement. The third harmonic supplies the difference when the two cosines equal one at $t = 0$.

$x(0) = 0.0996875 + 0.0003125 = R = 0.10 \text{ m}, \quad \dot{x}(0) = 0.$
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A good frequency prediction does not guarantee accurate phase for arbitrarily long times. An omitted $O(\varepsilon^2)$ frequency term can produce $O(\varepsilon^2\tau)$ phase drift.

2.4 Poincaré–Lindstedt method for cubic stiffness

Pendulum: recover the softening correction

Start with $\ddot{\theta} + (g/L) \sin \theta = 0$. With $\theta = Ru$ and $\tau = \sqrt{g/L} t$, divide by $R(g/L)$:

$$\theta = Ru, \quad \tau = \sqrt{g/L} t, \quad u'' + \frac{\sin(Ru)}{R} = 0.$$

Expand $\sin(Ru) = Ru - R^3 u^3/6 + O(R^5)$ and divide by R .

$$u'' + u - \frac{R^2}{6} u^3 + O(R^4) = 0.$$

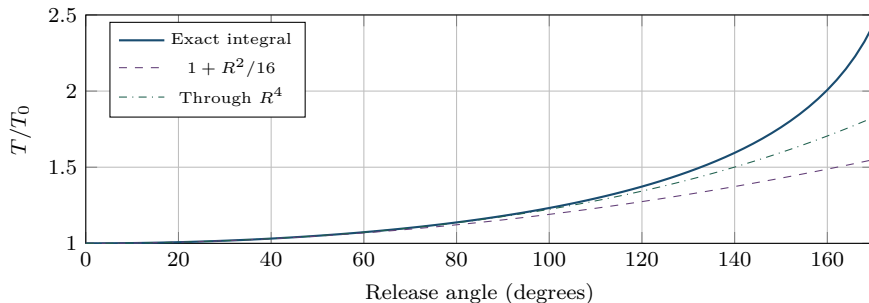
Here $\varepsilon = -R^2/6$. Insert it into $\Omega = 1 + 3\varepsilon/8 + O(\varepsilon^2)$.

$$\varepsilon = -R^2/6, \quad \boxed{\omega/\omega_0 = 1 - R^2/16 + O(R^4)}.$$

At $R = 30^\circ$, PL gives 0.9828653; the exact-model ratio is 0.9828891. This agrees with the energy-integral expansion. Use radians.

2.4 Poincaré–Lindstedt method for cubic stiffness

Pendulum: know when the small-angle expansion fails



Near $R = \pi$, the exact period diverges. To obtain the correct R^4 frequency term, retain both the next perturbation order and the quintic sine term.

2.5 Quadratic stiffness: mean shift and frequency

Why can the left and right turns differ?

In $m\ddot{x} + kx + k_2x^2 = 0$, the term k_2x^2 has the same sign on either side of $x = 0$. For $k_2 > 0$ and small $|x|$, the restoring force is stronger at $x > 0$ and weaker at $x < 0$.

$$x = Ru, \quad \tau = \sqrt{k/m} t, \quad \delta = \frac{k_2 R}{k}.$$

Substitute $\ddot{x} = R(k/m)u''$ and divide the equation by kR :

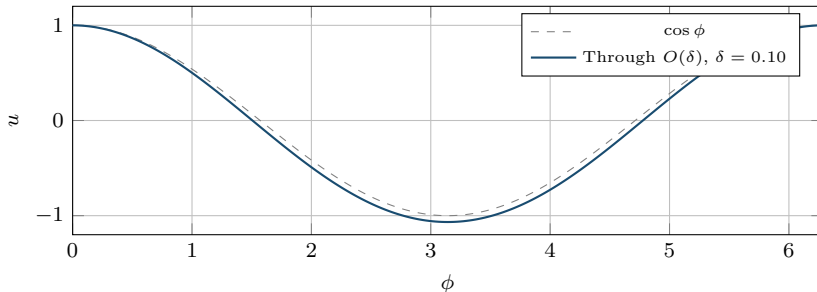
$$\boxed{u'' + u + \delta u^2 = 0}, \quad u(0) = 1, \quad u'(0) = 0.$$

Here R is the positive release from rest, primes mean derivatives in τ , and $|\delta| \ll 1$ for the perturbation calculation that follows.

2.5 Quadratic stiffness: mean shift and frequency

Quadratic stiffness makes a cycle asymmetric

$$u'' + u + \delta u^2 = 0, \quad \delta = 0.10, \quad u(0) = 1, \quad u'(0) = 0.$$



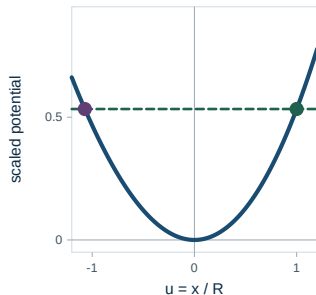
This is displacement u against phase $\phi = \Omega\tau$ over one cycle, not a phase-plane plot. The turning points are $+1$ and about -1.072 ; the mean is about -0.05 .

2.5 Quadratic stiffness: mean shift and frequency

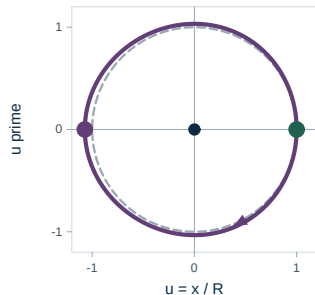
Potential and phase orbit show the same asymmetry

For $u'' + u + \delta u^2 = 0$ with $\delta = 0.10$, release at $u = 1$ gives the scaled potential $V(u) = u^2/2 + \delta u^3/3$ and energy level $V(1) = 1/2 + \delta/3$.

Energy sets unequal turns



Phase orbit: left turn -1.0718



The energy line and phase orbit share turning points $u = 1$ and $u = -1.0718$. Equilibrium remains at $u = 0$. The negative mean found next is the time average of this asymmetric cycle.

2.5 Quadratic stiffness: mean shift and frequency

Expand the phase and waveform

For the scaled model $u'' + u + \delta u^2 = 0$, use the phase $\phi = \Omega\tau$ and expand both waveform and frequency:

$$u = u_1 + \delta u_2 + \delta^2 u_3 + \cdots, \quad \Omega = 1 + \delta\Omega_1 + \delta^2\Omega_2 + \cdots.$$

As in the cubic case, $d^2u/d\tau^2 = \Omega^2 u_{\phi\phi}$. Insert the expansions and collect the terms independent of δ :

$$\Omega^2 u_{\phi\phi} + u + \delta u^2 = 0, \quad O(1): u_{1,\phi\phi} + u_1 = 0 \implies u_1 = \cos \phi.$$

The leading term already satisfies release, so $u_2(0) = u_{2,\phi}(0) = 0$.

2.5 Quadratic stiffness: mean shift and frequency

The first correction has no resonant term

Collect the δ terms in $\Omega^2 u_{\phi\phi} + u + \delta u^2 = 0$:

$$u_{2,\phi\phi} + u_2 = -u_1^2 - 2\Omega_1 u_{1,\phi\phi}.$$

Since $u_1 = \cos \phi$, use $u_{1,\phi\phi} = -\cos \phi$ and $\cos^2 \phi = (1 + \cos 2\phi)/2$:

$$u_{2,\phi\phi} + u_2 = 2\Omega_1 \cos \phi - \frac{1}{2}(1 + \cos 2\phi).$$

The quadratic term gives a constant and $\cos 2\phi$, but no $\cos \phi$. Set $2\Omega_1 = 0$ so the right side has no resonant $\cos \phi$ term.

$$\boxed{\Omega_1 = 0}, \quad u_{2,\phi\phi} + u_2 = -\frac{1}{2} - \frac{1}{2} \cos 2\phi.$$

Quadratic stiffness changes the waveform at first order. Its frequency shift begins at the next order.

2.5 Quadratic stiffness: mean shift and frequency

Solve the first correction: mean shift and second harmonic

$$u_{2,\phi\phi} + u_2 = -\frac{1}{2}(1 + \cos 2\phi).$$

A constant $-1/2$ solves the constant part. Since $(\cos 2\phi)_{\phi\phi} + \cos 2\phi = -3\cos 2\phi$, the second-harmonic part is $\cos 2\phi/6$.

Add a free $C \cos \phi + S \sin \phi$. At release, $0 = u_2(0) = -1/2 + 1/6 + C$ and $0 = u_{2,\phi}(0) = S$; thus $C = 1/3$, $S = 0$.

$$u_2 = -\frac{1}{2} + \frac{1}{3} \cos \phi + \frac{1}{6} \cos 2\phi.$$

The mean is $\bar{u} = -\delta/2 + O(\delta^2)$. At $\phi = \pi$, $u_2 = -2/3$, so the negative turning point is $u_- = -1 - 2\delta/3 + O(\delta^2)$.

2.5 Quadratic stiffness: mean shift and frequency

The frequency shift appears at second order

Now $\Omega_1 = 0$ and $\Omega^2 = 1 + 2\delta^2\Omega_2 + \dots$. The δ^2 terms give

$$u_{3,\phi\phi} + u_3 = -2u_1u_2 - 2\Omega_2u_{1,\phi\phi}.$$

Insert $u_1 = \cos \phi$ and $u_2 = -1/2 + (1/3)\cos \phi + (1/6)\cos 2\phi$. Product-to-sum gives

$$-2u_1u_2 = \frac{5}{6}\cos \phi - \frac{1}{3} - \frac{1}{3}\cos 2\phi - \frac{1}{6}\cos 3\phi.$$

Because $u_{1,\phi\phi} = -\cos \phi$, the frequency term adds $2\Omega_2 \cos \phi$. Remove the resonant cosine by setting

$$2\Omega_2 + \frac{5}{6} = 0 \quad \Longrightarrow \quad \boxed{\Omega = 1 - \frac{5}{12}\delta^2 + O(\delta^3)}.$$

The leading frequency contribution is softening for either sign of k_2 . It comes from feedback of the first-order waveform.

2.5 Quadratic stiffness: mean shift and frequency

Worked example: what does $\delta = 0.10$ predict?

$$u'' + u + 0.10u^2 = 0, \quad u(0) = 1, \quad u'(0) = 0.$$

From the preceding calculations, use the first-order waveform for displacement and the second-order correction for frequency:

$$\Omega_{\text{PL}} = 1 - \frac{5}{12}(0.10)^2 = 0.99583333, \quad \bar{u} \approx -\frac{0.10}{2} = -0.0500.$$

At the negative turning point $\phi = \pi$, $u_1 = -1$ and $u_2 = -2/3$, so

$$u_- \approx -1 + 0.10(-2/3) = -1.066667.$$

Energy quadrature gives $u_- = -1.071797$ and $\Omega = 0.995498$. The approximation captures the mean shift and softening; its turning-point error reflects omitted higher waveform orders.

2.6 Combined quadratic and cubic stiffness

The two nonlinear terms enter at different orders

Can a spring with positive cubic stiffness still soften? Use the same release R and the same scaled displacement $u = x/R$ as before:

$$m\ddot{x} + kx + k_2x^2 + k_3x^3 = 0, \quad \tau = \omega_0 t, \quad \omega_0 = \sqrt{k/m}.$$

Divide by kR and reuse the two dimensionless strengths already introduced:

$$\boxed{u'' + u + \delta u^2 + \varepsilon u^3 = 0}, \quad \delta = \frac{k_2 R}{k}, \quad \varepsilon = \frac{k_3 R^2}{k}.$$

As R becomes small, δ is proportional to R while ε is proportional to R^2 . The quadratic term first changes the waveform; its feedback and the cubic term change frequency at order R^2 .

2.6 Combined quadratic and cubic stiffness

Find the frequency change from the resonant cosine

From 2.5, $u_1 = \cos \phi$ and $u_2 = -1/2 + (1/3) \cos \phi + (1/6) \cos 2\phi$. The correction at order R^2 has three contributions to $\cos \phi$:

$$\cos \phi \cos 2\phi = \frac{1}{2}(\cos 3\phi + \cos \phi) \implies [u_1 u_2]_{\cos \phi} = -\frac{1}{2} + \frac{1}{12} = -\frac{5}{12}.$$

$$\text{Quadratic feedback: } -2\delta^2 u_1 u_2 \longrightarrow +\frac{5}{6}\delta^2 \cos \phi,$$

$$\text{Cubic term: } -\varepsilon u_1^3 \longrightarrow -\frac{3}{4}\varepsilon \cos \phi,$$

$$\text{Frequency change: } -2(\Omega - 1)u_{1,\phi\phi} \longrightarrow +2(\Omega - 1) \cos \phi.$$

A periodic correction needs the sum of these resonant coefficients to vanish:

$$2(\Omega - 1) + \frac{5}{6}\delta^2 - \frac{3}{4}\varepsilon = 0 \implies \boxed{\Omega \approx 1 + \frac{3}{8}\varepsilon - \frac{5}{12}\delta^2.}$$

The cubic contribution has the sign of k_3 . The quadratic contribution is negative for either sign of k_2 .

2.6 Combined quadratic and cubic stiffness

Frequency and displacement in physical units

Insert $\delta = k_2 R/k$ and $\varepsilon = k_3 R^2/k$ into $\omega = \omega_0 \Omega$:

$$\omega \approx \omega_0 \left[1 + \left(\frac{3k_3}{8k} - \frac{5k_2^2}{12k^2} \right) R^2 \right].$$

For the displacement, retain the first quadratic correction from 2.5. The cubic correction to the waveform is one order smaller in R :

$$x \approx R \cos \phi + \frac{k_2 R^2}{k} \left(-\frac{1}{2} + \frac{1}{3} \cos \phi + \frac{1}{6} \cos 2\phi \right), \quad \phi \approx \omega t.$$

At $\phi = 0$ the correction sums to zero, so $x(0) = R$. Averaging the cosines gives $\bar{x} \approx -k_2 R^2/(2k)$.

The frequency formula retains terms proportional to R^2 ; the displayed waveform retains displacement terms through R^2 .

2.6 Combined quadratic and cubic stiffness

Worked example: positive cubic stiffness, net softening

$$m = 1 \text{ kg}, \quad k = 100 \text{ N/m}, \quad k_2 = 200 \text{ N/m}^2, \quad k_3 = 100 \text{ N/m}^3, \quad R = 0.050 \text{ m}.$$

Here $\omega_0 = \sqrt{100/1} = 10 \text{ rad/s}$, $\delta = 200(0.050)/100 = 0.10$, and $\varepsilon = 100(0.050)^2/100 = 0.0025$.

$$\frac{3}{8}\varepsilon = 0.00093750, \quad -\frac{5}{12}\delta^2 = -0.00416667.$$

$$\omega \approx 10(1 - 0.00322917) = 9.9677083 \text{ rad/s}.$$

Quadratic feedback outweighs positive cubic stiffness at this release. The first waveform correction gives $\bar{x} \approx -0.002500 \text{ m}$ and $x_- \approx -R(1 + 2\delta/3) = -0.053333 \text{ m}$.

2.6 Combined quadratic and cubic stiffness

When do the leading stiffness effects cancel?

The coefficient of R^2 in the frequency formula determines the initial direction of the backbone:

$$\frac{3k_3}{8k} > \frac{5k_2^2}{12k^2} : \text{hardening}, \quad \frac{3k_3}{8k} < \frac{5k_2^2}{12k^2} : \text{softening}.$$

At the boundary, set the two terms equal. Solving for k_3 and inserting the example's $k = 100$ and $k_2 = 200$ gives

$$k_3 = \frac{10k_2^2}{9k} = \frac{10(200)^2}{9(100)} = 444.444 \text{ N/m}^3.$$

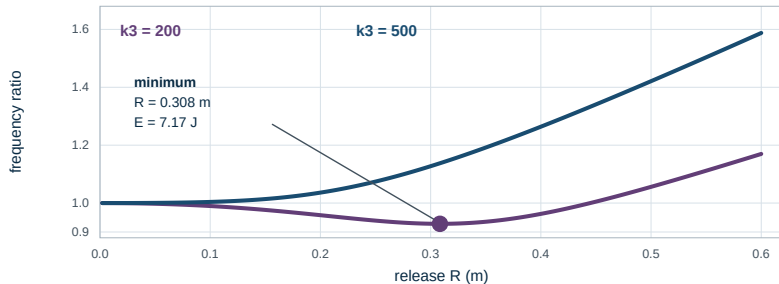
Cancellation of the R^2 term does not prove that every amplitude has the same frequency. Higher-order terms remain.

2.6 Combined quadratic and cubic stiffness

Can the backbone change direction as energy rises?

Keep $m = 1$ kg, $k = 100$ N/m and $k_2 = 200$ N/m²; compare $k_3 = 200$ and 500 N/m³. For each release, set $E = U(R)$ and use the exact energy-period integral from 2.1.

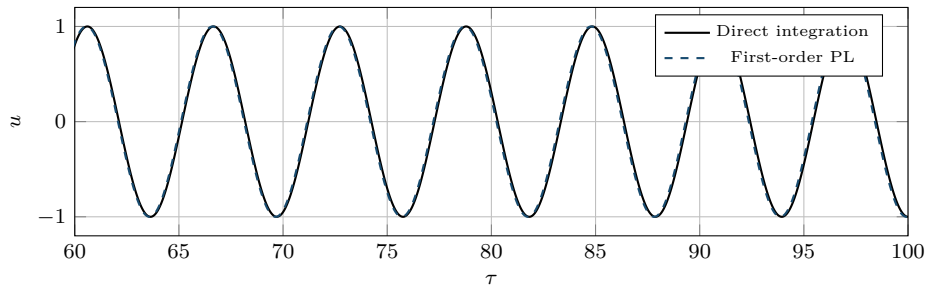
Exact free-orbit backbones



For $k_3 = 200$, frequency first falls to $\omega/\omega_0 = 0.928$ at $R = 0.308$ m ($E = 7.17$ J), then rises as energy increases. For $k_3 = 500$ it rises over the plotted range. The R^2 formula predicts only the initial direction.

2.7 Validation, limitations and solved practice

Can a small frequency error matter?



For $\varepsilon = 0.10$, $\Delta\Omega = 0.000783093$. By $\tau = 100$, the phase difference is approximately 0.0783093 rad. PL does not remove errors from every omitted order.

2.7 Validation, limitations and solved practice

Validate frequency against an independent reference

For the same cubic model and release $u(0) = 1$, $u'(0) = 0$, evaluate the energy-period integral from 2.2 numerically, then use $\Omega_{\text{quad}} = 2\pi/T_\tau$. Compare it with $\Omega_{\text{PL}} = 1 + 3\varepsilon/8$.

ε	Quadrature Ω	First-order PL	Error (%)
0.02	1.00746750	1.00750000	0.003226
0.10	1.03671691	1.03750000	0.075536
0.30	1.10601824	1.11250000	0.586045
-0.10	0.96163744	0.96250000	0.089697
-0.30	0.87878975	0.88750000	0.991164

Error is $100|\Omega_{\text{PL}} - \Omega_{\text{quad}}|/\Omega_{\text{quad}}$. Both calculations use the same cubic model, scaled time and release R .

2.7 Validation, limitations and solved practice

Check the period, conservation and asymptotic order

Three independent checks

1. Integrate $u' = v$, $v' = -u - \varepsilon u^3$. Consecutive downward crossings of $u = 0$ are one period apart. Refine the time step.

$$\mathcal{E} = \frac{1}{2}v^2 + \frac{1}{2}u^2 + \frac{1}{4}\varepsilon u^4, \quad \Delta_E = \max_{\tau} \frac{|\mathcal{E}(\tau) - \mathcal{E}(0)|}{|\mathcal{E}(0)|}.$$

2. Compare the measured period with converged energy quadrature; track energy drift throughout the record.
3. Substitute the PL waveform into the equation. A first-order cubic approximation should leave a residual of order ε^2 at fixed phase.

If the residual is $C\varepsilon^2 + O(\varepsilon^3)$, replacing ε by $\varepsilon/2$ gives approximately one quarter as much residual.

A factor near four. For $0.01 \rightarrow 0.005$, the calculated ratio is 4.00214. Residual, frequency error, energy drift and long-time phase error measure different things.

2.7 Validation, limitations and solved practice

Solved practice: a softening oscillator

$$m = 2 \text{ kg}, \quad k = 50 \text{ N/m}, \quad \alpha = -200 \text{ N/m}^3, \quad R = 0.10 \text{ m}.$$

Scale the spring model as in 2.4. Its linear frequency and nonlinear strength are

$$\omega_0 = \sqrt{50/2} = 5 \text{ rad/s}, \quad \varepsilon = \frac{-200(0.10)^2}{50} = -0.0400.$$

The frequency ratio is $\Omega \approx 1 + 3(-0.0400)/8 = 0.9850$, so

$$\omega \approx 5(0.9850) = 4.925 \text{ rad/s}.$$

In the release-preserving waveform, $\varepsilon/32 = -0.00125$ and $\phi = \omega t$:

$$x \approx 0.10[\cos \phi - 0.00125(\cos 3\phi - \cos \phi)] \text{ m}, \quad \phi = 4.925t.$$

Fundamental: 0.100125 m; signed third harmonic: -0.000125 m . Their sum is R .

2.7 Validation, limitations and solved practice

Softening practice: check the energy barrier

$$U(x) = \frac{1}{2}kx^2 - \frac{1}{4}|\alpha|x^4.$$

The nonzero stationary points solve $U'(x) = x(k - |\alpha|x^2) = 0$, hence $x_s = \pm\sqrt{k/|\alpha|}$. Compare their potential energy with the release energy $E = U(R)$.

$$x_s = \pm\sqrt{k/|\alpha|} = \pm 0.50 \text{ m},$$

$$U(x_s) = \frac{k^2}{4|\alpha|} = \frac{50^2}{4(200)} = 3.125 \text{ J},$$

$$E = U(0.10) = \frac{1}{2}(50)(0.10)^2 - \frac{1}{4}(200)(0.10)^4 = 0.245 \text{ J}.$$

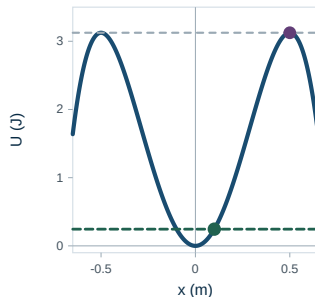
The release lies inside the central well and below the barrier. The potential is not globally confining, but this small central oscillation is bounded. Near a separatrix, small-amplitude perturbation is not reliable.

2.7 Validation, limitations and solved practice

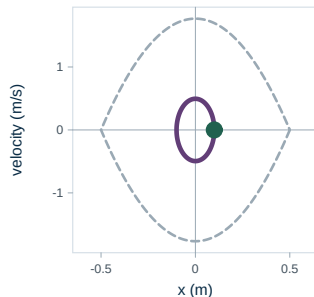
The softening example stays inside its energy barrier

For the preceding example, $E = 0.245$ J at $R = 0.10$ m and the barrier is 3.125 J at $x = \pm 0.50$ m.

Release energy below barrier



Small central orbit



Green marks release energy and the purple central orbit. Gray marks the barrier and its phase contour. This recalls Week 1; the Week 2 frequency estimate describes the small bounded orbit.

2.7 Validation, limitations and solved practice

Exit check: what should we be able to explain?

Why no first-order quadratic frequency shift?

The square of the leading cosine contains a mean and a second harmonic, but no fundamental.

Why does quadratic stiffness change frequency later?

The first-order waveform feeds a fundamental harmonic into the next-order equation.

What removes the secular term?

Choose the frequency correction so that the resonant Fourier coefficient vanishes.

What must be checked before applying a formula?

The release R , smallness of the nonlinear terms, model truncation and the time interval of interest.

Next week: weak damping, slowly varying amplitude and phase, multiple scales and averaging.

WEEK 2 REFERENCES

Sources for the methods and examples

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Nonlinear Oscillations. Wiley, 1979.

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[2] **O. Thomas**

“Understanding, computing and identifying the nonlinear dynamics of elastic and piezoelectric structures thanks to nonlinear modes.” In *Model Order Reduction for Design, Analysis and Control of Nonlinear Vibratory Systems*, 2024.

Fig. 4.4: the pendulum’s free-oscillation backbone and its changing waveform; the driven-response curves in that figure serve as a comparison. DOI: 10.1007/978-3-031-67499-0_4.